

```

.MODEL SMALL
.STACK 100H
.DATA
    msg DB 'SHL Output:',0DH,0AH,'$'
    nl   DB 0DH,0AH,'$'
    num DB 1
.CODE
MAIN PROC
    MOV AX,@DATA
    MOV DS,AX
    MOV AH,9
    LEA DX,msg
    INT 21H
    MOV CX,5
L1:
    MOV AL,num
    SHL AL,1
    ADD AL,'0'
    MOV DL,AL
    MOV AH,2
    INT 21H
    MOV AH,9
    LEA DX,nl
    INT 21H
    INC num
    LOOP L1
    MOV AH,4CH
    INT 21H
MAIN ENDP
END MAIN

```

```

-----

.MODEL SMALL
.STACK 100H
.DATA
    msg DB 'SHR Output:',0DH,0AH,'$'
    nl   DB 0DH,0AH,'$'
    num DB 8
.CODE
MAIN PROC
    MOV AX,@DATA
    MOV DS,AX
    MOV AH,9
    LEA DX,msg
    INT 21H
    MOV CX,4

```

L2:

```
    MOV AL,num
    SHR AL,1
    ADD AL,'0'
    MOV DL,AL
    MOV AH,2
    INT 21H
    MOV AH,9
    LEA DX,nl
    INT 21H
    SHR num,1
    LOOP L2
    MOV AH,4CH
    INT 21H
```

MAIN ENDP

END MAIN

.MODEL SMALL

.STACK 100H

.DATA

```
    msg DB 'RCL Output:',0DH,0AH,'$'
    nl   DB 0DH,0AH,'$'
    num DB 3
```

.CODE

MAIN PROC

```
    MOV AX,@DATA
    MOV DS,AX
    CLC
    MOV AH,9
    LEA DX,msg
    INT 21H
    MOV CX,4
```

L3:

```
    MOV AL,num
    RCL AL,1
    ADD AL,'0'
    MOV DL,AL
    MOV AH,2
    INT 21H
    MOV AH,9
    LEA DX,nl
    INT 21H
    INC num
    LOOP L3
    MOV AH,4CH
```

```

        INT 21H
MAIN ENDP
END MAIN

-----

.MODEL SMALL
.STACK 100H
.DATA
    msg DB 'ROR Output:',0DH,0AH,'$'
    nl  DB 0DH,0AH,'$'
    num DB 8
.CODE
MAIN PROC
    MOV AX,@DATA
    MOV DS,AX
    MOV AH,9
    LEA DX,msg
    INT 21H
    MOV CX,4
L4:
    MOV AL,num
    ROR AL,1
    ADD AL,'0'
    MOV DL,AL
    MOV AH,2
    INT 21H
    MOV AH,9
    LEA DX,nl
    INT 21H
    INC num
    LOOP L4
    MOV AH,4CH
    INT 21H
MAIN ENDP
END MAIN

```